

When 3% means nothing: calibrating escalation limits to a bank's own forecasting error distribution.

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Motivation

NII FORECASTING AND IRRBB GOVERNANCE

- Forecast accuracy for net interest income (NII) and interest-rate risk in the banking book (IRRBB) is central to earnings stability, balance-sheet management and supervisory credibility.
- Many institutions govern forecast performance with fixed deviation thresholds (3/4/5%), even though forecast uncertainty widens with the horizon and the error process may be heavy-tailed. A flat limit therefore carries no consistent probabilistic interpretation.
- The paper develops an integrated, probability-coherent framework for monitoring NII forecast errors and assessing IRRBB limit breaches, calibrated to the bank's own empirical error distribution.

| Where forecast errors come from

NII FORECAST CONSTRUCTION

- NII baselines are built under run-off, static or dynamic balance-sheet assumptions, combined with an interest-rate path (market forwards or house views) and pricing and behavioral inputs. Each assumption is a source of forecast error.
- The paper distinguishes sensitivities, responses of NII to prescribed shocks, from risk measures, statements about the probability distribution of outcomes. Escalation limits belong to the second category.
- The governance angle is distributional and horizon-aware: limits are calibrated to the empirical forecast-error distribution rather than attributing each miss causally.

Horizon-specific, quantile-anchored thresholds

THE SEP FAN-CHART LOGIC APPLIED TO ESCALATION LIMITS

Following the Federal Reserve's RMSE-based communication of forecast uncertainty, thresholds are anchored to the horizon-specific error scale:

$$c_{h,q} = z_q \cdot RMSE(h), \quad z_{80} = 0.84, \quad z_{90} = 1.28, \quad z_{95} = 1.64, \quad z_{99} = 2.33$$

under the Normal/RMSE mapping used for communication, with empirical quantiles replacing $z_q RMSE(h)$ where the error history warrants it.

- Escalation tiers then inherit a coverage interpretation: a q -tier limit is expected to be exceeded with probability $1 - q$ at every horizon, which no flat limit can deliver.

A common unit across horizons

STANDARDIZED EXCEEDANCE SCORE

Each realized relative error $e_{t,h}$ is expressed as a proportion of the 95% boundary for its horizon,

$$S_{t,h} = \frac{|e_{t,h}|}{Q_h(0.95)},$$

so that dashboards report one number with the same meaning at every h .

- $S_{t,h} < 1$ lies inside the 95% band, $S_{t,h} \geq 1$ marks an exceedance, and the magnitude above one grades its severity.
- The score separates the monitoring object (a standardized statistic) from the governance object (the tier boundaries), so recalibration of limits does not change the reporting format.

The outer tier: extreme value theory

POT-GPD CALIBRATION OF RARE MISSES

- Because NII forecast errors may be heavy-tailed, the outer escalation tier is calibrated by peaks-over-threshold: by the Pickands–Balkema–de Haan theorem, excesses over a high threshold converge to a Generalized Pareto Distribution.
- In the empirical illustration the EVT overlay is applied only to the outer tier, under the $\xi \rightarrow 0$ exponential-tail approximation with diagnostics, to stabilise deep tails estimated from few observations.
- The construction is analogous to Expected-Shortfall extensions of the Basel VaR traffic-light backtest: probabilistic coverage targets operationalised into governance tiers.

Simulated thresholds by horizon

N=120 PER HORIZON · 5 EXCEEDANCES AT THE 95% THRESHOLD

Empirical thresholds of $|e_{t,h}|$ (relative NII forecasting errors) and the EVT outer tier:

	$h = 3m$	$h = 6m$	$h = 12m$	$h = 24m$
80%	0.99%	1.40%	1.96%	2.81%
90%	1.25%	1.74%	2.43%	3.46%
95%	2.01%	2.82%	4.00%	5.54%
EVT 97.5%	2.63%	3.73%	5.29%	7.36%

Average exceedances $\hat{\beta}_h$ rise with the horizon as well: 1.21%, 1.79%, 2.52% and 3.55%. Dispersion growth with h is embedded in the limit definition itself.

Twelve limits versus one flat number

HORIZON-SPECIFIC SYSTEM VS FLAT 3/4/5%

- The 12-limit system (three tiers at four horizons) aligns exceedance rates across horizons: every tier means the same probability everywhere.
- Against flat 3/4/5% limits the comparison shows fewer false alarms at long horizons, where flat limits over-penalise the naturally wider errors, and fewer missed tails at short horizons, where flat limits are too loose.
- Backtesting of the system is clean in the simulation, with threshold stability demonstrated across replications.

Statistical validation

COVERAGE, ELICITABILITY, DECOMPOSITION

- Interval-forecast evaluation follows Christoffersen (1998): unconditional coverage, whether the long-run frequency of exceedances matches the nominal level, and conditional coverage, whether exceedances are independent rather than clustered.
- Kupiec's proportion-of-failures test (1995) provides the simpler unconditional benchmark familiar from the 1996 Basel traffic-light framework for VaR backtesting.
- Quantile elicibility justifies the quantile as the reported functional, and a bias-dispersion decomposition separates structural misspecification from random variation in the error process.

The IRRBB extension

BREACH PROBABILITIES FOR Δ NII LIMITS

- The methodology extends to IRRBB by quantifying the probability that limits on Δ NII are breached solely due to forecast or model uncertainty, with no change in the underlying risk position.
- This maps model risk into risk-appetite language: a limit whose pure-noise breach probability is material is not measuring the intended exposure, which matters under the Basel IRRBB Standard (BCBS 368).
- Breach probabilities per horizon are directly suitable for ALCO reporting alongside the exceedance score.

Limitations and adoption path

WHAT THE FRAMEWORK REQUIRES

- Calibration is data-hungry at long horizons: the 24-month tier rests on few effective observations, which the EVT overlay mitigates but does not remove.
- Regime or model changes can break stationarity of the error process, requiring re-baselining and shadow backtests.
- At adoption the paper recommends wider bands or pooled-horizon estimation until the institution's own error history is long enough, with a sequenced implementation road map.

| Implications

FOR FORECAST GOVERNANCE

- Escalation limits gain a coverage interpretation: management knows the probability content of every tier at every horizon, instead of a flat number that means a different probability at each h .
- Comparability across horizons is restored because dispersion growth is embedded in the limit definition, and conditional-coverage testing detects clustering of exceptions.
- The same architecture serves NII forecast monitoring and IRRBB limit design, giving one consistent statistical language for both.

Thank you.

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